



Novel nuclear power and fossil fuel hybrid propulsion systems for long-endurance unmanned aerial vehicles: Configuration comparison and performance optimization

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ABSTRACT

Aircraft nuclear propulsion system can significantly enhance the endurance, but its large size and weight become disadvantages against the conventional fossil fuel propulsion system. In this paper, a novel nuclear power and fossil fuel hybrid propulsion system, which combines the lithium-cooled fast reactor and the kerosene-fueled turbine engine, is proposed to meet the power requirements of different flight segments. Performance assessment and comparison among different hybrid configurations is performed. Results indicate that the configuration combining the nuclear power module with a high bypass ratio turbofan engine is the most promising for aircraft. The primary optimization shows that there is a trade-off between the weight of the nuclear power module and the specific thrust, and the optimal weight of nuclear power module is 3942 kg, with the specific thrust of 195.8 N kg⁻¹ s⁻¹. The aircraft equipped with the hybrid propulsion system operates at the maximum altitude of approximately 18.2 km and reaches a maximum Mach number of 0.71, allowing for sustained flight for several months in cruise segment. The turbofan engine, nuclear reactor, ancillary systems, shield, and heat exchanger account for 11 %, 23 %, 11 %, 17 % and 1 % of the fuel mass of the original unmodified aircraft.

Nomenclature

C_D	Drag coefficient	22	Outlet of fan inner
C_{Dmin}	Drag coefficient at the minimum lift	13	Outlet of fan culvert
C_L	Lift coefficient	25	Inlet of compressor
C_{Lmin}	Minimum lift coefficient	18	Outlet of bypass nozzle
F	Thrust (N)	3	Outlet of compressor
K	lift induced drag factor	31	Inlet of burner
M	Weight (kg)	4	Outlet of burner
P	Pressure (pa)	41	Inlet of HPT
Q	Heat transfer power (W)	44	Outlet of HPT
S	Area (m ²)	45	Inlet of LPT
T	Temperature (K)	5	Outlet of LPT
c_p	Constant pressure specific heat (J·kg ⁻¹ ·K ⁻¹)	8	Outlet of core nozzle

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f	Fuel-to-air ratio	c1	Cold side inlet of heat exchanger
h	Specific enthalpy (J·kg ⁻¹)	c2	Cold side outlet of heat exchanger
$\dot{m}$	Mass flow rate (kg·s ⁻¹)	h1	Hot side inlet of heat exchanger
n	speed (rpm)	h2	Hot side outlet of heat exchanger
q	Free stream dynamic pressure	in	Inlet
v	Velocity (m·s ⁻¹)	out	Outlet
γ	Ratio of Specific Heat	ANP	Aircraft nuclear propulsion
ϵ	Effectiveness of heat exchanger	BPR	Bypass ratio
η	Efficiency	HPT	High pressure turbine

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(continued)

π	Pressure ratio	HWR	Heat-to-weight ratio
σ	Pressure recovery coefficient	LPT	Low pressure turbine
Subscript		OPR	Overall pressure ratio
0	Inlet of engine	T.I.T.	Turbine inlet temperature
1	Inlet of fan	UAV	Unmanned aerial vehicle

1. Introduction

High-altitude long endurance (HALE) unmanned aerial vehicle (UAV), due to its flexible deployment capabilities and the ability to establish communication links with the ground over a wide area, has significant advantages in Earth observation tasks such as remote sensing, space-ground communication, weather forecasting, wildfire monitoring, and military reconnaissance [1]. The fossil fuel propulsion systems, which are utilized as power units for UAVs, face challenges in achieving endurance times exceeding two days due to limitations in range and endurance. Compared to traditional fossil fuel propulsion systems, the aircraft nuclear propulsion system has the advantage of carrying heavy loads and achieving extremely long endurance [2]. From the 1950s to the 1960s, the United States initiated the Aircraft Nuclear Propulsion (ANP) program aimed at researching aircraft nuclear propulsion systems. The project was canceled due to the inability to produce the reactor that was both safe and controllable, while also meeting the power requirements of flight at a sufficiently low weight [3]. Over the past 40 years, little research has been conducted specifically related to nuclear powered aircraft engines. With the increasing demand for greater power unit, new and advanced propulsion systems are crucial for meeting long-term technological needs. Recent developments in the field of improved reliability and high inherent safety advantages of fast reactors [4,5], as well as strategies for reducing risks in the event of a reactor crash (such as utilizing Spectral Shift Absorbers (SSAs) to counteract reactivity increases) [6] have significantly increased the potential for using nuclear power to drive aircraft. Additionally, advances in airframe design [7,8], airframe manufacturing technology [9], jet engine technology [10–12], and materials [13] further enhance this possibility.

With the support of nuclear energy, the nuclear propulsion system has significantly reduced the amount of fuel carried [14]. Mitrica et al. [15] gradually conducted relevant tests on the nuclear power system and preliminary calculations of the dynamic cycle characteristics. Bai et al. [16] proposed a novel multi-annular structure nuclear heat exchanger, which placed in the combustion chamber to heat compressed air. For the nuclear-powered aircraft, the mission will not last for several years, so its lifetime is not the primary consideration. It requires the high-power density output of the nuclear reactor to help the aircraft meet the demands of high-thrust operating segments. Under the given value of the output thrust, the minimum weight of the reactor could provide the system with the best dynamic response performances. The cruise thrust is further lower than the takeoff thrust for most of the aircraft [17,18], and they mostly work in the cruise segment [19]. A lower thermal power reactor can provide enough cruise power for the aircraft, but they cannot provide enough takeoff power. However, the reasonable propulsion system should take cruise and takeoff segments into account. Therefore, combining the use of different power units to leverage their performance and advantages becomes a solution. Hamilton et al. [20] discussed the possibility of the dual-energy engine for UAVs. Engine mode changeover operation from the conventional combustion to the isomer energy source is described. However, the authors did not propose a specific layout scheme for the conventional combustion and isomer energy source in the engine.

In summary, this paper considers establishing a novel hybrid propulsion system that combines the advantages of high-energy density

from nuclear power and high-power density from fossil fuels. Under cruise segment, the nuclear reactor operates independently, with its thermal power only needing to meet the power requirements of the high-altitude operation. Consequently, the reactor can be designed relatively small, resulting in significant fuel savings, while the weight penalty imposed by the reactor remains minimal. During takeoff segment, when the burner is operational, both the specific power and specific energy of the system increase, enabling the UAV to perform well under both takeoff and cruise segments, which differs from previous aircraft nuclear propulsion systems. Therefore, this hybrid propulsion system represents a propulsion solution suitable for long-endurance UAVs, capable of meeting the power and energy requirements throughout the entire flight mission. Currently, the turbine engines are widely used in long-endurance UAVs due to their high power characteristics, mostly in the form of turbofan engines [21,22] and turbojet engines [23,24]. Turbofan engines are highly efficient and commonly used in most commercial and civilian aircraft [25,26]. Turbojet engines have slightly lower efficiency than turbofan engines but perform well at high speeds, mainly used in some military aircraft and specific high-speed applications [27,28].

The models of hybrid propulsion system that combines different types of turbine engines with a small lithium-cooled reactor are first established in this paper. The performance and weight distribution of different configurations of the hybrid propulsion system in various flight segments is investigated, and comparing them with traditional turbofan engines. Subsequently, the configuration that meets the performance requirements is selected, and the influence of component parameters on the performance of the novel hybrid propulsion system is studied, with optimization being carried out for two parameters: the weight of the nuclear power module and the specific thrust. Finally, based on the optimized component parameters, the flight envelope and weight distribution are being presented to demonstrate the prospects and limitations of the hybrid propulsion system.

2. System description

The configuration of the hybrid propulsion system is shown in Fig. 1. The entire system can be considered as consisting of a turbine engine module and a nuclear power module (a loop composed of the nuclear reactor and the heat exchanger). In order to meet design requirements, the reactor portion employs a small modular lithium-cooled fast reactor, positioned inside the unmanned aerial vehicle (UAV) compartment, while the heat exchanger is located within the engine. High-temperature lithium coolant is pumped from the reactor outlet to the heat exchanger, where heat is exchanged between air and lithium. Finally, the lithium exiting the heat exchanger re-enters the reactor to complete the cycle. The closed-loop heat exchange method effectively avoids the risk of nuclear fuel leakage associated with an open-loop system [29].

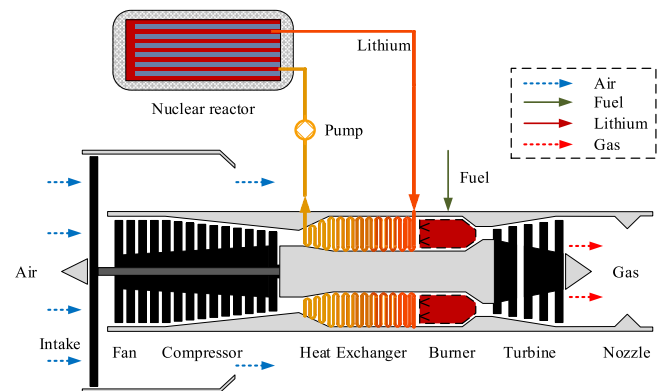


Fig. 1. Scheme diagrams of the nuclear power and fossil fuel hybrid propulsion system.

Additionally, the use of lithium as a heat transfer medium, with a working temperature exceeding 1500 K [30], significantly enhances the efficiency of the power system, achieving a greater thrust-to-weight ratio. Simultaneously, a portion of the tanks of the UAV is retained, meeting the thrust requirements during takeoff and climb segments.

Under the cruise segment, the nuclear power mode is employed. The air entering the engine inlet is compressed by components such as the fan and compressor before entering the heat exchanger. Within the heat exchanger, heat is transferred from lithium to air through either conduction or convective heat transfer. The burner is closed, the high-temperature air drives the turbine and output propulsion power in the nozzle. Under takeoff and climb segments, the reactor is in shutdown mode, and fuel is injected into the burner, where it operates normally. The high-temperature gases exiting the burner and output propulsion power in the nozzle. The operating status of rotating components such as the fan and compressor vary across different flight segments. The pressure ratio and mass flow of components such as the fan and compressor change with rotational speed.

3. Mathematic modeling

3.1. Aircraft model

The selection of appropriate engine operating conditions necessitates a deep understanding of the aerodynamic properties of the aircraft in which the engine is installed. This study focuses on the investigation of the HALE UAV. This vehicle is anticipated to operate at high altitude and relatively low Mach number. The assumed properties for this aircraft are summarized in Table 1. Thrust is assumed to be equal to drag, which is calculated using standard methodologies [31]. The assumed form of the drag polar is presented in Eqs. (1)–(4).

$$C_D = C_{Dmin} + K(C_L - C_{Lmin})^2 \quad (1)$$

$$C_L = \text{Lift} / (qS) \quad (2)$$

$$C_D = \text{Drag} / (qS) \quad (3)$$

$$q = \rho_{\text{air}} v_0^2 / 2 \quad (4)$$

where C_L is the lift coefficient, C_D is the drag coefficient, q is the free-stream dynamic pressure, v_0 is the flight speed, and S is the wing area. C_{Lmin} is the minimum lift coefficient, and C_{Dmin} is the drag coefficient at C_{Lmin} .

3.2. Thermodynamic model

The novel hybrid propulsion system is consisted of the turbine engine module (including compressor, turbine and burner) and nuclear power

module (including nuclear reactor, pump and heat exchanger). The following general assumptions are used in this analysis.

- 1) The hybrid propulsion system is simulated for steady-state operation;
- 2) Zero leakage occurs in the hybrid propulsion system;
- 3) Gaseous working fluids are considered as ideal gases;
- 4) Heat transfer, resistance and friction losses from lithium to the environment are negligible;
- 5) The aircraft can be operated with the Mach number from 0 to 1.

3.2.1. Turbine engine

In this paper, three different hybrid power systems are presented, and the performance of those is compared with that of the actual engine. The actual engine configuration and three novel configurations are.

- 1) Configuration 1: Turbofan, high bypass ratio (BPR), separate exhaust, OPR = 24, BPR = 5;
- 2) Configuration 2: Nuclear-turbofan, high BPR, separate exhaust, OPR = 24, BPR = 5;
- 3) Configuration 3: Nuclear-turbofan, low BPR, combined exhaust, OPR = 24, BPR = 0.8;
- 4) Configuration 4: Nuclear-turbojet, OPR = 24.

The component efficiencies are chosen to be consistent with late 20th century technology [32], and Details of the engine parameters are presented in Table 2.

The specific aerospace thermodynamic component-level modeling process of the separated-exhaust turbofan configuration is shown in Fig. 2. In this configuration, an additional heat exchanger is added between the compressor and the burner. The aerothermodynamic parameters such as pressure, temperature, and flow rate of each component are calculated in turn along the airflow process, and then, the steady-state calculations of the component-level model are realized by combining the equilibrium equations that must be satisfied by each component to work together [33].

(1) Inlet

The intake is considered adiabatic in the model, a typical total pressure recovery coefficient is adopted in the model as shown in Eq. (5).

$$\sigma_{in} = \begin{cases} 1 & , Ma \leq 1 \\ 1 - 0.075 \times (Ma - 1)^{1.35} & , Ma > 1 \end{cases} \quad (5)$$

The total temperature and pressure at the intake outlet are shown in Eq. (6).

$$\begin{cases} T_1^* = T_0^* \\ P_1^* = \sigma_{in} P_0^* \end{cases} \quad (6)$$

(2) Fan and compressor

The expression of the relative corrected speed is shown in Eq. (7).

Table 1
Simulated aircraft specifications [31].

Parameter	Value	Units
Airframe		
Wingspan, b	40.0	m
Wing areas, S	64.0	m ²
Loaded airframe	7000	kg
Fuel capacity	7500	kg
Aerodynamics		
C_{Dmin}	0.0195	
K	0.01725	
C_{Lmin}	0.3	
Power plant		
Number of engines	1	
Total rated thrust	≥ 35.0	kN
Operation		
Altitude	16.8	km
Mach number	0.6	

Table 2
Performance parameters related to the turbine engine [32].

Parameter	Value	Units
Fan efficiency, η_{Fan}	0.86	
Compressor efficiency, η_C	0.88	
Combustion efficiency, η_b	0.99	
Burner pressure recovery coefficient, σ_b	0.98	
T.I.T	≤ 1560	K
HPT efficiency, η_{HPT}	0.87	
LPT efficiency, η_{LPT}	0.87	

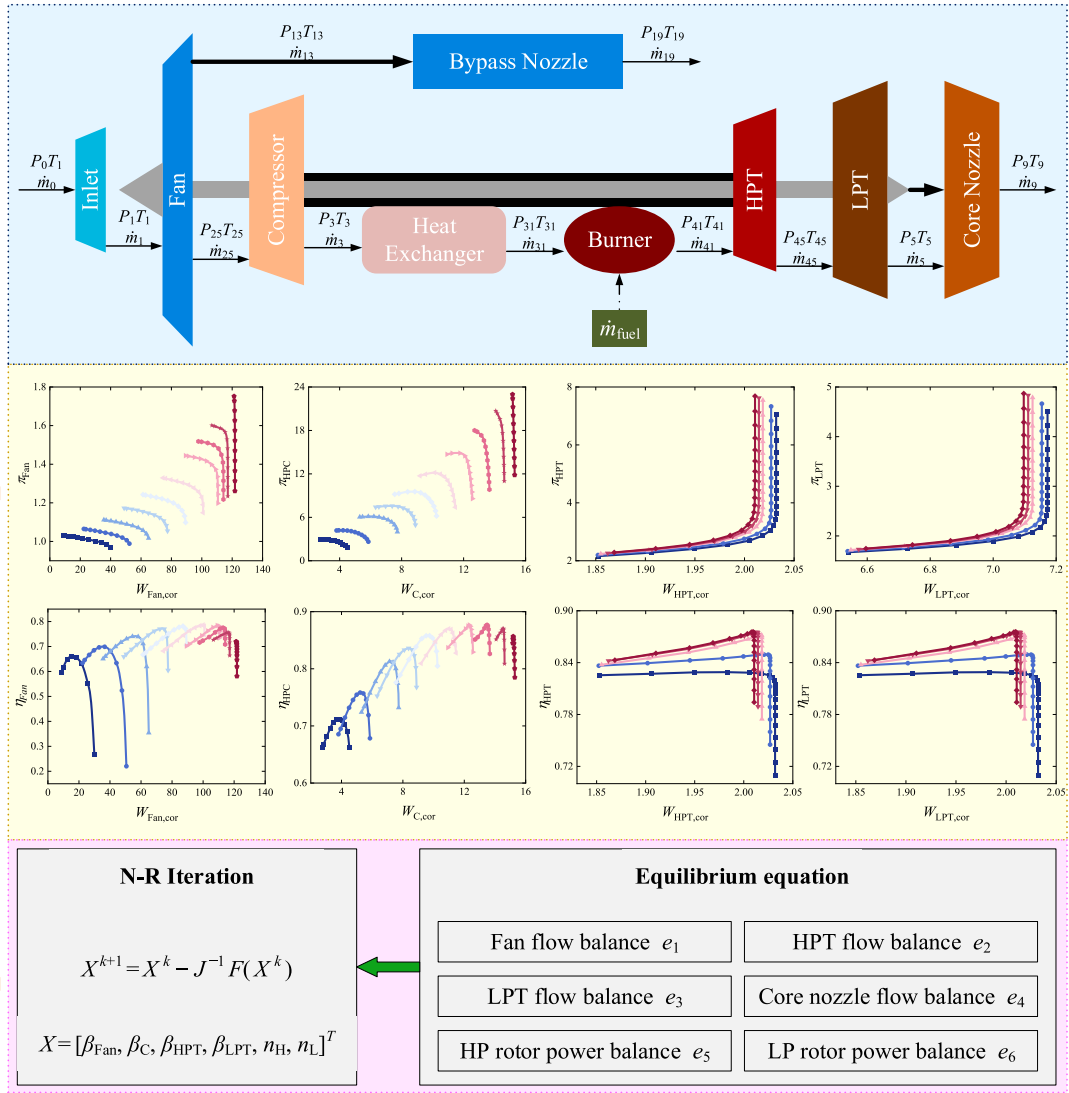


Fig. 2. Working principle of the separated-exhaust turbofan engine configuration.

$$\bar{n}_{cor} = \frac{n/\sqrt{T_{in}}}{n_d/\sqrt{T_{in,d}}}$$

(7)

where the subscript “d” represents the designing parameter.

The pressure ratio and efficiency of the fan and compressor is the function of the corrected shaft speed and mass flow [34].

$$\begin{cases} \pi = f_\pi(\bar{n}_{cor}, \dot{m}_{cor}) \\ \eta = f_\eta(\bar{n}_{cor}, \dot{m}_{cor}) \end{cases} \quad (8)$$

The total pressure and temperature of the air at the outlet can be calculated as follows:

$$\begin{cases} P_{out}^* = P_{in}^* \cdot \pi \\ T_{out}^* = T_{in}^* \left(1 + \frac{\gamma-1}{\eta_c} \pi^{\frac{\gamma-1}{\gamma}} \right) \end{cases} \quad (9)$$

(3) Burner

The total pressure, mass flow rate, fuel-air ratio, outlet total enthalpy, and total temperature of the burner can be calculated as follows [35]:

$$P_{out}^* = P_{in}^* \sigma_b \quad (10)$$

$$f_{out} = \frac{\dot{m}_{fuel}}{\dot{m}_{in}} \quad (11)$$

$$h_{out}^* = \frac{\dot{m}_{fuel} H_{VF} \eta_b + \dot{m}_{in} h_{in}^*}{\dot{m}_{in} + \dot{m}_{fuel}} \quad (12)$$

$$T_{out}^* = f_T(f_{out}, h_{out}^*) \quad (13)$$

where f is the fuel-to-air ratio, H_{VF} is the calorific value of the fuel, σ_b is the total pressure recovery coefficient, η_b is the combustion efficiency.

(4) Turbine

The corrected flow rate and isentropic efficiency can be calculated as follows [34]:

$$\begin{cases} \dot{m}_{cor} = f_m(n_{cor}, \pi) \\ \eta = f_\eta(n_{cor}, \pi) \end{cases} \quad (14)$$

The total temperature at the outlet of the turbines is as shown in Eq. (15).

$$T_{out}^* = T_{in}^* \left(1 - \left(1 - \pi^{\frac{\gamma-1}{\gamma}} \right) \eta_T \right) \quad (15)$$

(5) Nozzle

The nozzle of the turbofan engine is convergent-divergent, after considering the loss of nozzle, the actual velocity at the exit of the nozzle is as shown in Eq. (16).

$$v = \varphi_{NZ} \sqrt{2(h_{out}^* - h_{out})} \quad (16)$$

3.2.2. Nuclear power system

The reactors can be designed with either a thermal or a fast neutron spectrum, and the size of the reactor was determined primarily by the reactor type [36]. Currently, there is limited research on reactor systems designed specifically for aircraft propulsion, with space applications primarily utilizing liquid metals and inert gases as coolants. Among these, lithium stands out due to its higher specific heat capacity compared to lead-bismuth alloy, allowing for a more compact core design and enabling operation at temperatures around 1600 K. Additionally, the lower density of lithium [37] makes it a lighter coolant, beneficial for weight-sensitive applications like aviation. While high-temperature gas-cooled reactors often combine primary and secondary cooling loops, this approach may not be feasible for air-breathing propulsion systems, potentially limiting their effectiveness in terms of weight and volume. Notably, comparative studies, such as those from the European MEGAHIT project [38], demonstrate a significant weight advantage for lithium-cooled reactors, with component weights of 1.1 tons versus 4.0 tons for gas-cooled systems. This reinforces the potential of liquid metal cooling systems in aircraft applications. For this study, the lithium-cooled fast neutron spectrum reactors are selected [39], and the major design parameters of the nuclear power system are listed in Table 3.

A counter-flow heat exchanger is assumed for each of them because it has a better heat transfer performance than other types [40]. The schematic system diagram of nuclear power system is shown in Fig. 3. As the heat source of the system, the lithium cooled reactor heats air indirectly to the maximum circulating temperature using the heat exchanger.

The working processes of heat exchangers are usually described by the inlet temperatures and pressures of the heat exchanging fluids, as well as the so-called effectiveness of heat exchanger that is defined as the ratio of the actual transferred heat to the theoretical maximum transferred heat [41],

$$\epsilon_{HE} = \frac{Q_{act}}{Q_{max}} \quad (17)$$

In Eq. (21), the actual and the maximum transferred heat is defined respectively as:

$$Q_{max} = \min(\dot{m}_{li} c_{p,li}, \dot{m}_{air} c_{p,air}) \cdot (T_{h1} - T_{c1}) \quad (18)$$

Table 3

Design parameters of nuclear power system [39].

Parameter	Value	Units
Reactor		
Rated thermal power	≤4	MW
Life time	1	year
Fission fuel	UN	
Maximum fuel temperature	1650	K
Coolant type	Liquid lithium	
Heat exchanger		
Effectiveness of heat exchanger, ϵ_{HE}	0.85	
Cold side pressure recovery coefficient, σ_c	0.85	
Inlet temperature of the hot side, T_{h1}	1500	K

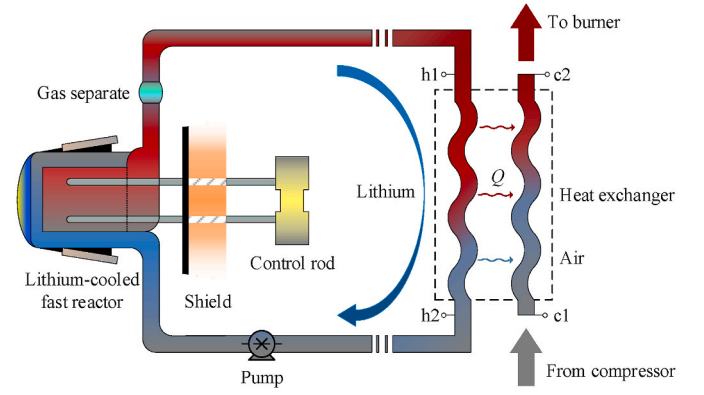


Fig. 3. Schematic diagram of the lithium-cooled fast reactor and counter-flow heat exchanger.

$$Q_{act} = \dot{m}_{li} c_{p,li} (T_{h1} - T_{h2}) = \dot{m}_{air} c_{p,air} (T_{c2} - T_{c1}) \quad (19)$$

The outlet pressures of the air can be obtained according to the inlet pressures and the pressure recovery factors.

$$P_{c2} = \sigma_c \cdot P_{c1} \quad (20)$$

3.3. Mass model

3.3.1. Turbine engine

The mass model used here is based on the work of Onat and Klees [42] who developed a methodology for predicting the mass of a gas turbine engine based on a detailed, piece-by-piece accounting of the entire engine configuration. According to the analysis of waters et al. [31], the average error is $\pm 14\%$. Therefore, in predicting the performance of hybrid systems, we believe that the quality model is sufficient for the comparisons made in this work.

3.3.2. Reactor

The mass of liquid-metal-cooled reactors can be estimated using the following formulas [43,44]. The reactor components comprise uranium fuel, structural elements, reflector, pressure vessel, and various miscellaneous components. The fuel mass is determined as the maximum of three values: the neutron critical mass, the specific power limiting mass, and the fuel damage limiting mass.

The total fuel mass based on neutronic limiting is the sum of the mass of uranium burned up and the mass of uranium at the end of the design life, which can be expressed as follows:

$$M_E = M_B + M_C \quad (21)$$

The total fuel mass based on specific power limits can also be expressed as determined by material-based thermal considerations and temperature limits, expressed as Eq. (22).

$$M_P = \frac{P_{th} P_F}{P_S} \quad (22)$$

where P_S is the maximum allow specific power, which can be expressed as Eq. (23):

$$P_S = \frac{T_f - T_c}{\rho_f R_T^T \times 10^6} \quad (23)$$

where T_f and T_c are the maximum fuel temperature and the core outlet bulk coolant temperature, respectively.

Fuel damage restricts the fraction of fuel available for burnup. Consequently, the fuel mass based on burnup limits for fuel damage is given by:

$$M_F = \frac{0.38P_{th}tP_F}{\beta_{Rea}} \quad (24)$$

The actual required fuel mass will be the largest of the fuel mass requirements based on neutronic, thermal, and fuel damage limits, which can be expressed as Eq. (25).

$$M_L = \max(M_E, M_P, M_F) \quad (25)$$

According to the fuel mass, the mass of the components including the structure, reflector, pressure vessel and coolant. The total mass of the reactor M_R is the sum of the masses of the components, which can be expressed as Eq. (26).

$$M_R = M_L + M_S + M_{RF} + M_{PV} + M_{LC} \quad (26)$$

The lithium electromagnetic pump in the primary circuit operates at high temperatures, where conventional ferromagnetic materials experience a rapid decline in magnetic permeability, significantly affecting pump efficiency. Currently, limited research exists on lithium electromagnetic pumps suitable for aircraft propulsion. This study adopts the ultra-high-temperature electromagnetic pump design developed by NASA [42] for liquid metal Rankine cycles in space applications. An annular linear electromagnetic pump is designed that incorporates cooling liquid from a secondary circuit to cool the electromagnetic coils and core, thereby reducing local temperatures and enabling the flow of lithium at 1422 K. The relevant parameters are shown in Table 4, and the specifications of the ultra-high-temperature electromagnetic pump meet the design requirements of this study.

The mass of the ancillary system includes instrumentation and control mass, safety system mass and electromagnetic pump mass, expressed as Eq. (27).

$$M_{AS} = M_{IC} + M_{SS} + M_{EMP} \quad (27)$$

where M_{IC} is the mass of the instrumentation and control (I&C); M_{SS} is the mass of the safety system; M_{EMP} is the mass of the electromagnetic pump.

3.3.3. Shield

The shadow shield is used to shield fast neutrons and gamma radial. The shielding effect can be obtained by the following estimation formulas [45].

$$A_n = \frac{D_n H M_U}{(1.57 \times 10^{20}) V P_{th} N_{U235}} \times \left(\ln \sqrt{R_p^2 + 0.75^2} - \ln R_p \right)^{-1} \quad (28)$$

$$A_g = \frac{D_g H}{(2 \times 10^7) P_{th}} \times \left(\ln \sqrt{R_p^2 + 0.75^2} - \ln R_p \right)^{-1} \quad (29)$$

where A_n and A_g are the neutron shielding factor and gamma shielding factor; D_n and D_g are the maximum allowable neutron dose and gamma dose on the load; H is the core height; P_{th} is the reactor thermal power; R_p is the distance between the load and the reactor; M_U is the Uranium loading in the core; V is the volume of the core; N_{U235} is the nuclear density of U235 in the core.

Referring to the shield layer arrangement proposed in SP-100 reach [46] of the United States, the dimension of each layer can be calculated through simple geometric relations. Beryllium (Be) is the thermal

conductivity material in the shield layout, tungsten (W) is the gamma shielding material, and LiH is the neutron shielding material. According to the volume and density of each layer, the total mass of the shielding layer can be expressed as:

$$M_S = \rho_{Be} V_{Be} + \rho_W V_W + \rho_{LiH} V_{LiH} \quad (30)$$

3.3.4. Heat exchanger

The weight of the heat exchanger is determined by its thermal load and the heat-to-weight ratio (HWR), which is influenced by factors such as the average temperature difference, materials, and manufacturing processes. For aviation heat exchangers, the HWR ranges from 30 to 45 kW kg⁻¹ [47]. Therefore, the mass of the heat exchanger can be expressed as:

$$M_{HE} = \frac{Q}{HWR} \quad (31)$$

The total weight of the nuclear power module is composed of the reactor, ancillary system, shield, and heat exchanger, and can be expressed as:

$$M_{NPM} = M_R + M_{AS} + M_S + M_{HE} \quad (32)$$

3.4. Performance parameters

For a hybrid propulsion system, its performance is usually evaluated by thrust, specific thrust, propulsion efficiency, and thermal efficiency [48], which are given in Eqs. (33)–(36).

$$F = \dot{m}_{core}[(1+f)v_9 - v_0] + \dot{m}_{Fan}(v_{19} - v_0) + A_9(P_9 - P_0) + A_{19}(P_{19} - P_0) \quad (33)$$

$$F_s = \frac{F}{\dot{m}_{air}} \quad (34)$$

$$\eta_p = \frac{Fv_0}{\dot{m}_{core}[(1+f)v_9^2 - v_0^2] + \dot{m}_{Fan}(v_{19}^2 - v_0^2)} \quad (35)$$

$$\eta_{th} = \frac{\dot{m}_{core}[(1+f)v_9^2 - v_0^2] + \dot{m}_{Fan}(v_{19}^2 - v_0^2)}{2(\dot{m}_{fuel}H_{VF} + Q_{input})} \quad (36)$$

In Eq. (36), when the burner in the hybrid power system is deactivated, the value of $\dot{m}_{fuel}$ is zero. Note that the thrust force of turbojet engines is all obtained from core section. Consequently, the performance calculations for turbojet engines in Eqs. (33)–(36) exclude the fan unit.

3.5. Solution strategy

Fig. 4 shows the solution strategy of the nuclear power and fossil fuel hybrid propulsion system. After determining the flight altitude and Mach number for the UAV cruise segment, the minimum thrust required to maintain the aircraft's equilibrium is firstly computed. Subsequently, the cruise segment is selected as the design point for the turbine engine, and parameters such as the mass flow rate of inlet air and heat exchanger are assumed to calculate the performance parameters of the turbine engine design point. Next, the minimum mass of the reactor module is determined based on the thermal power of the heat exchanger.

When operating at off-design points, the nuclear reactor ceases operation, necessitating the determination of the working condition of the turbine engine. A series of independent variables determine the working state, and an equal number of equilibrium equations are established accordingly. These equilibrium equations are based on principles of mass conservation and physical constraints such as flow and power balance. All these equations form a set of nonlinear equations. The Newton-Raphson algorithm [49] is employed to find solutions of Eq. (37), implemented in Python, and is renowned for its fast convergence and high precision. Convergence is considered achieved

Table 4
Design characteristics of annular linear electromagnetic pump [42].

Parameter	Value	Units
Fluid velocity	8.74	m/s
Duct wall thickness	0.0016	m
Duct cross section	0.1092 (od)	m
Fluid passage	0.1062 (od) × 0.0808 (id)	m
Approximate total mass	326.6	kg
Overall dimensions	0.3048 (od) × 1.4224 (long)	m

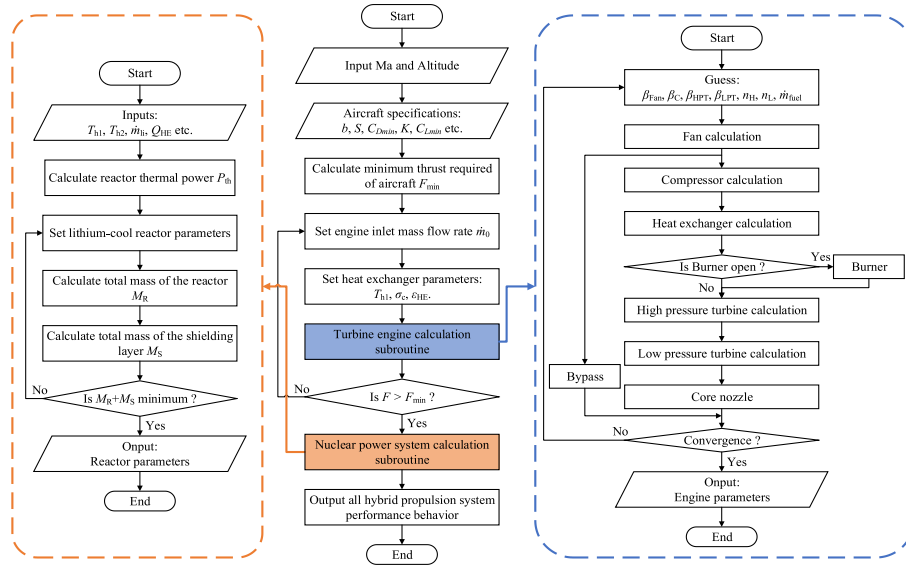


Fig. 4. Solution strategy block diagram.

when the maximum value on left-hand-side if Eq. (37) is smaller than 1×10^{-10} . When all equilibrium equations are satisfied, the working conditions are uniquely determined, allowing for further determination of the state of each component and the overall performance of the turbine engine.

$$\begin{cases} e1 = (\dot{m}_1 - \dot{m}_8 - \dot{m}_{18})/\dot{m}_1 \\ e2 = (\dot{m}_4 - \dot{m}_{41})/\dot{m}_4 \\ e3 = (\dot{m}_{45} - \dot{m}_5)/\dot{m}_{45} \\ e4 = (\dot{m}_{51} - \dot{m}_8)/\dot{m}_{51} \\ e5 = (\dot{W}_C - \eta_{HP} \dot{W}_{HPT})/\dot{W}_C \\ e6 = (\dot{W}_{Fan} - \eta_{LP} \dot{W}_{LPT})/\dot{W}_{Fan} \end{cases} \quad (37)$$

4. Results and analysis

4.1. Model validation of turbine engine

To ensure the accuracy of the turboprop engine thermodynamic model, the simulation results of this model was compared with those of Ref. The modeling object is a dual-spool, separated-flow turboprop engine. The experimental results are given in the literature [50] and the turboprop engine model established in this study were compared. The working condition is set to be on the maximum take-off thrust ($Ma = 0$, $H = 0$ km). The simulation results of the model are shown in Table 5. Under the same flight conditions, the relative error of all performance parameters is less than 2 %. Besides, the engine mass model is also verified by the in-serve turboprop engine published data [19], and the results have been compared, in which the relative errors do not exceed 4 %, as shown in Table 6. Therefore, the turboprop engine model can be

Table 5
Turboprop engine thermodynamic model verification.

Parameters	Values		
Flight altitude (km)	0		
Mach number	0		
Mass flow rate of air ($\text{kg}\cdot\text{s}^{-1}$)	35.58		
Mass flow rate of fuel ($\text{kg}\cdot\text{s}^{-1}$)	0.177		
Results	Ref. [50]	This paper	Relative error
The temperature of bypass nozzle (K)	330.15	330.02	-0.04 %
The pressure of bypass nozzle (kPa)	154.71	154.69	-0.01 %
The velocity of bypass nozzle (m/s)	265.07	262.86	-0.87 %
T.I.T (K)	1283.15	1276.31	-0.53 %
Inlet temperature of LPT (K)	990.45	993.52	+0.31 %
Thrust (kN)	11.24	11.04	-1.78 %

Table 6
Turboprop engine mass model verification.

Parameters	Values		
Bypass ratio	4.79		
OPR	23		
Thrust (kN)	42.23		
Specific fuel consumption ($\text{g}\cdot\text{kN}^{-1}\cdot\text{s}^{-1}$)	11.03		
Results	Ref. [19]	This paper	Relative error
Weight (kg)	746	722	-3.22 %

considered reliable.

4.2. Performance comparisons of hybrid propulsion system configurations

A typical flight mission for the aircraft generally consists of five distinct segments: take-off, climb, cruise, descent, and landing, as illustrated in Fig. 5. There are significant regulatory issues and environmental concerns relating to operation of a high-power, gamma-ray sources. Therefore, restrictions have been imposed on the operation of nuclear reactors, requiring them to operate at altitudes above 20,000 ft (6.1 km) [14], while fossil fuel is used below this altitude. In this study, it is assumed that the reactor is started during the cruise segment (at an altitude of 16.8 km), and after adjusting to the desired power level, the hybrid propulsion system switches from fossil fuel to nuclear power. The nuclear power system operates only during the cruise segment,

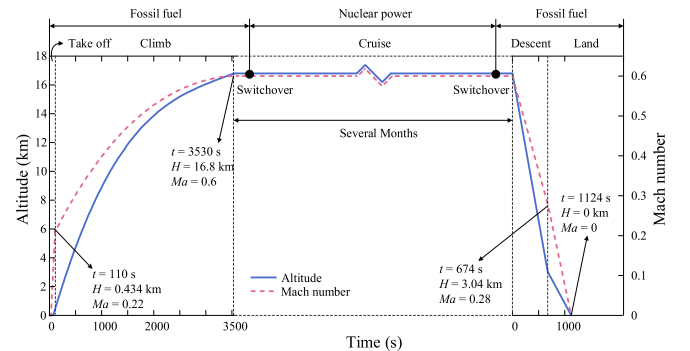


Fig. 5. Hybrid propulsion system working profile after being simplified reference UAV [51,52].

continuously releasing energy through nuclear fission to meet the UAV cruise requirements for months. Similarly, before the UAV lands, the power source switches to fossil fuel and shuts down the reactor. The reactor power only needs to meet the UAV flight requirements during the cruise segment, with its size and power determined by this segment. The maximum take-off weight and total rated thrust of the UAV are determined in Table 1.

When calculating off-design points for the hybrid propulsion system, technical indicators for each flight segment must be considered. To this end, a simplified flight mission table is provided, as shown in Table 7. This table includes nine mission points: take-off, climb 1, climb 2, climb 3, climb 4, cruise, descent 1, descent 2, and landing. Each mission point is accompanied by relevant data such as altitude, flight Mach number, turbine inlet temperature, burner operating conditions, and flight time.

4.2.1. Cruise segment

The cruise phase, which accounts for a significant portion of flight time, especially for long-duration flights, it holds crucial importance for most aircraft. Therefore, this paper considers employing nuclear power propulsion system to enhance the endurance of UAVs. According to Fig. 6, the thrust required to balance the aircraft is approximately 4.2 kN with the altitude and Mach number of the aircraft are 16.8 km and 0.6. Fig. 7 illustrates the performance comparison of different configuration hybrid propulsion systems during the cruise segment under the parameters provided in Table 7. The specific thrust of different engine configurations during the cruise segment is shown in Fig. 7 (a). Compared to configuration 1, configuration 2 exhibits slightly lower specific thrust due to additional total pressure loss caused by the heat exchanger, despite having the same bypass ratio and overall pressure ratio. For configuration 3 with a smaller bypass ratio and configuration 4 with the turbojet engine configuration, they demonstrate higher specific thrust. Fig. 7 (b) and (c) show the engine air mass flow and diameter, respectively. With the same engine thrust, a lower specific thrust implies a need for the larger air mass flow. However, the larger engine diameter introduces greater aerodynamic drag during flight, which is unfavorable for weight reduction in hybrid propulsion systems. Fig. 7 (d) illustrates the engine thermal efficiency and propulsion efficiency. Considering the additional pressure loss from the heat exchanger, the thermal efficiency of configurations 2, 3 and 4 is lower than that of configuration 1. With other parameters unchanged, reducing the engine bypass ratio leads to decrease in the overall engine thermal load, thereby improving thermal efficiency. However, some of the gas kinetic energy cannot be converted into thrust, resulting in decrease in propulsive efficiency.

4.2.2. Take-off, climb, descent, and land segments

The tracks of the aircraft during take-off, climb, descent, and land segments are complex. Based on real flight missions as reference [51], we have simplified them and presented the work status of each flight segment in Table 7. The total rated thrust requirement for the take-off segment is greater than 35 kN, as shown in Table 1. The thrust in different flight segments is illustrated in Fig. 8 (a), showing that the thrust generated by the hybrid propulsion system is higher than the

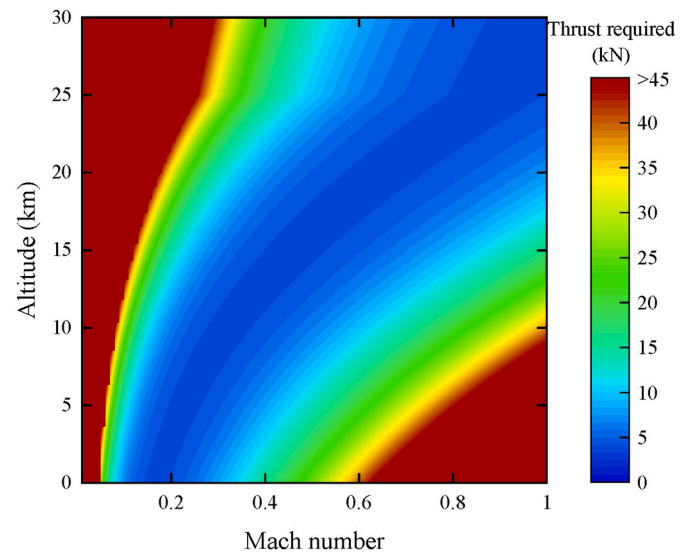


Fig. 6. Thrust needed to balance the aircraft.

thrust required, thus all three configurations of the hybrid propulsion system can meet the thrust requirement of the aircraft. Among them, the thrust generated by the hybrid propulsion system in climb 1 and 2 greatly exceeds the required thrust, enabling the aircraft to have a higher climb rate. The variations of air and fuel mass flow rate in different flight segments are shown in Fig. 8 (b) and (c). With the increase in flight altitude, the air density decreases, leading to the reduction in the amount of air ingested by the engine during the climb segment, thus reducing the required fuel mass. Configurations 1, 2 and 3 need to adjust the fan speed and operating state to meet different thrust requirements in different flight segments, resulting in significant differences in air mass flow. On the other hand, configuration 4 mainly adjust fuel supply and burner operating conditions rather than fan speed. Therefore, their variation in air mass flow is relatively stable across different flight segments, but fuel mass flow rate varies greatly. During climb, as flight altitude increases, the environmental temperature decreases, resulting in the increase in the thermal efficiency of the hybrid propulsion system shown in Fig. 8 (d). Conversely, during descent and land, as flight altitude decreases, the thermal efficiency decreases accordingly.

Fig. 9 presents the weight distribution of various components in the hybrid propulsion system. Configuration 1 consists only of the engine weight (730 kg), while configurations 2, 3 and 4 include the weights of the engine, reactor, ancillary system, shield, and heat exchanger, with total system component weights of 5437 kg, 6435 kg, and 6997 kg, respectively. It can be observed that the weights of the components rank from largest to smallest as reactor (42 %), shield (26 %), engine (14 %), ancillary system (16 %), and heat exchanger (2 %). Due to its high energy density, the reactor can meet the energy demands of long-duration flights, but its lower power density results in the larger weight.

The fuel consumption comparison of the hybrid propulsion system in different flight segments is illustrated in Fig. 10. Among them, the fuel consumed during the climb segment is the highest because the aircraft needs to overcome gravity and attain sufficient altitude during this phase, requiring greater thrust. Additionally, the climb segment lasts the longest, necessitating the engine to burn more fuel. Configuration 4 exhibits lower efficiency, resulting in the highest total fuel consumption across different flight segments (1900 kg), followed by configurations 3, 2 and 1, with 1764 kg, 1261 kg, and 1254 kg respectively.

Fig. 9 illustrates the component weights of the hybrid propulsion system, while Fig. 10 shows the fuel weight to complete the mission. To ensure the UAV can take off smoothly, the sum of the component weight and the carried fuel weight of the hybrid propulsion system must be less

Table 7

Hybrid propulsion system working status of complete flight mission envelope point.

Flight Mission	Altitude (km)	Mach number	Burner	T.I.T (K)	Time (s)
Take-off	0	0	ON	1540	110
Climb 1	0.434	0.22	ON	1535	240
Climb 2	3.2	0.27	ON	1495	600
Climb 3	8.5	0.39	ON	1423	600
Climb 4	12	0.48	ON	1404	1980
Cruise	16.8	0.6	OFF	1370	–
Descent 1	12	0.48	ON	1404	240
Descent 2	8.2	0.4	ON	1427	434
Land	3.04	0.28	ON	1498	450

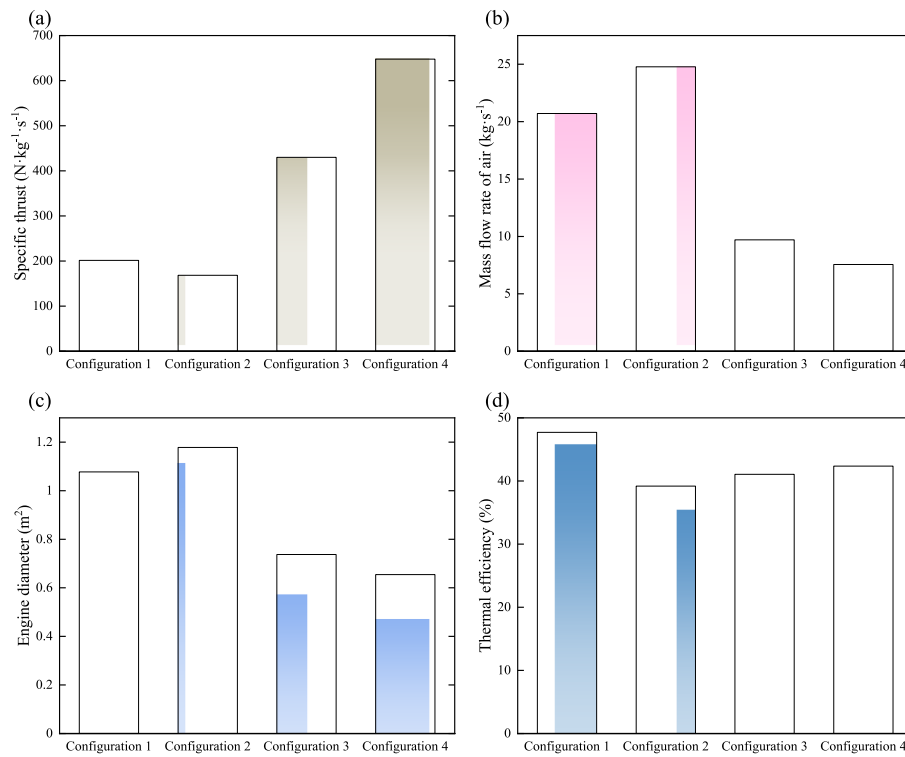


Fig. 7. Performance comparison of the hybrid power system in the cruise segment (a) specific thrust; (b) mass flow rate of air; (c) engine diameter; (d) thermal efficiency.

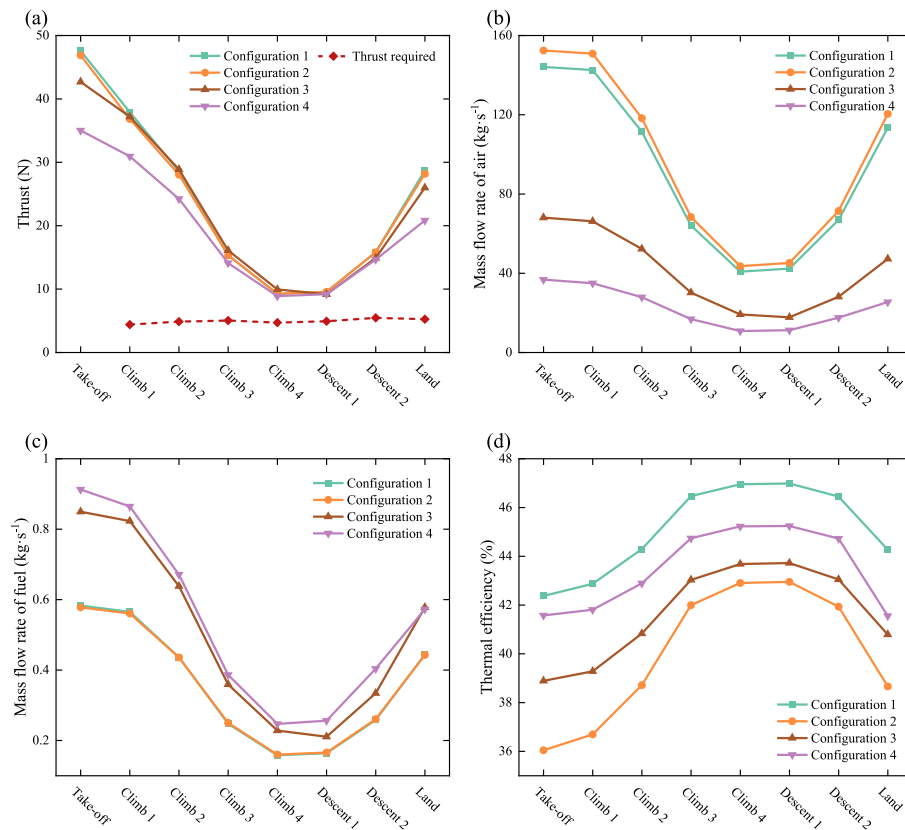


Fig. 8. Performance comparison of the hybrid power system in the take-off, climb, descent, and land segments (a) thrust; (b) mass flow rate of air; (c) mass flow rate of fuel; (d) thermal efficiency.

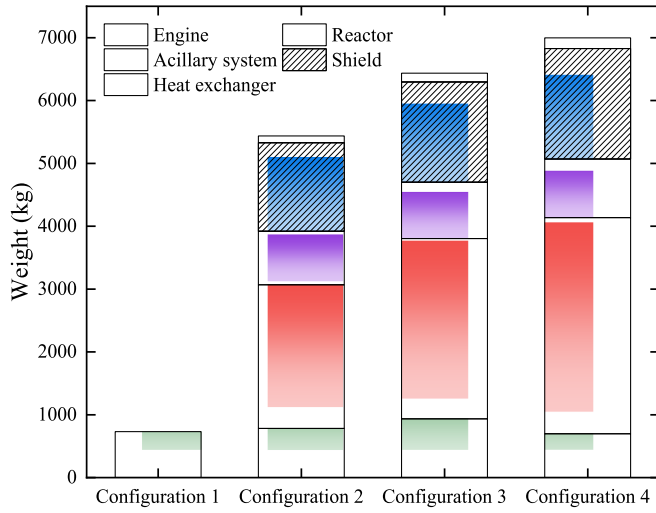


Fig. 9. Weight distribution of various components in the hybrid propulsion system.

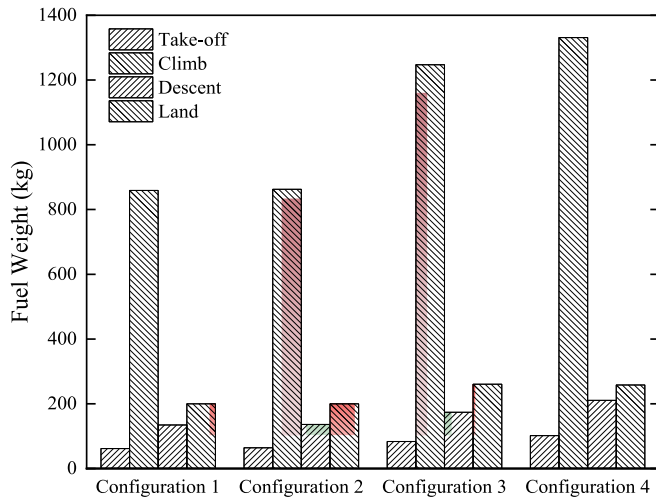


Fig. 10. Hybrid propulsion system fuel consumption comparison in different flight segments.

than the UAV fuel capacity provided in Table 1. However, for configurations 3 and 4, their total weights are 8199 kg and 8897 kg respectively, both exceeding the maximum weight limit. Therefore, the hybrid propulsion system plans to adopt the configuration similar to configuration 2, featuring high BPR separate exhaust turbofan engine design.

4.3. Performance analysis of hybrid power system for separate-exhaust turbofan engine configuration

4.3.1. Sensitivity analysis

A sensitivity analysis is performed to quantify the impact of several important parameters on the performance of the hybrid propulsion system which are bypass ratio (BPR), fan pressure ratio (π_{Fan}), compressor pressure ratio (π_C), inlet temperature of the hot side of the heat exchanger (T_{h1}), and the effectiveness of heat exchanger (ϵ_{HE}). Table 8 presents the sensitivity coefficients, defined as the ratio of the fractional change in system performance parameter to the fractional change in disturbance parameters.

Increasing the bypass ratio of a turbofan engine can enhance propulsion efficiency but results in a decrease in specific thrust. Raising the π_{Fan} increases the power required by the fan, thereby elevating the

Table 8
Sensitivity analysis.

ΔY	Weight	Specific thrust	Thermal efficiency	Propulsion efficiency
Symbols	M_{NPM}	F_s	η_{th}	η_p
BPR ± 1	-0.16	-0.55 %	-0.14 %	+0.42 %
%	%			
	+0.16	+0.56 %	+0.14 %	-0.42 %
%	%			
$\pi_{Fan} \pm 1$	-0.34	+0.54 %	-0.12 %	+0.84 %
%	%			
	+0.35	-0.57 %	+0.11 %	-0.86 %
%	%			
$\pi_C \pm 1$	-0.09	+0.02 %	+0.21 %	-0.03 %
%	%			
	+0.09	-0.03 %	-0.22 %	+0.03 %
%	%			
$T_{h1} \pm 1$	+3.07	+1.05 %	+0.43 %	-1.07 %
%	%			
	-2.77	-1.08 %	-0.46 %	+1.11 %
%	%			
$\epsilon_{HE} \pm 1$	+0.18	+0.62 %	+0.26 %	-0.63 %
%	%			
	-0.18	-0.63 %	-0.27 %	+0.64 %
%	%			

overall thermal load of the engine. Consequently, although the fan can augment both specific thrust and propulsion efficiency, the thermal efficiency of the system tends to decrease due to more thermal energy being converted into shaft power rather than thrust. In contrast, the impact of the π_C on system performance is relatively minor. Compared to other parameters, the T_{h1} exerts a greater influence on the weight of the nuclear power module (NPM). For the reactor, when reactor thermal power and core fuel temperature remain constant, an increase in the coolant outlet temperature leads to a reduction in the temperature difference between the core and the coolant, thereby increasing the volume and weight of the reactor. For engine performance, increasing T_{h1} can elevate the turbine inlet temperature, aiding in enhancing both specific thrust and thermal efficiency. The heat transfer effectiveness of the heat exchanger is also crucial for specific thrust and thermal efficiency, although it may result in a decrease in propulsion efficiency.

4.3.2. Influence of bypass ratio and fan pressure ratio on system performance

In this section, the influence of bypass ratio and fan pressure ratio on the performance of the hybrid propulsion system is studied, in which the range of bypass ratio is from 2 to 10 and the range of fan pressure ratio is from 1.1 to 1.9, while other influencing factors remain at the design conditions. Fig. 11 illustrates the system performance affected by bypass ratio and fan pressure ratio. When the bypass ratio is relatively low, increasing the fan pressure ratio results in an increase in engine specific thrust and a decrease in the weight of the nuclear power system module. However, in scenarios with higher bypass ratios, increasing the fan pressure ratio leads to excessive conversion of thermal energy into shaft work, causing a decrease in engine specific thrust and an increase in the weight of the nuclear power system module, as shown in Fig. 11 (a) and (b). Overall, increasing the bypass ratio can reduce the weight of the nuclear power system module. The thermal efficiency and propulsion efficiency are displayed in Fig. 11 (c) and (d). While increasing the bypass ratio can enhance propulsion efficiency, it also elevates the thermal load of the system, thereby the thermal efficiency decreases.

4.3.3. Influence of OPR and T_{h1} on system performance

In this section, the effects of overall pressure ratio (OPR) and the inlet temperature of the hot side of the heat exchanger (T_{h1}) on the performance of the hybrid propulsion system are studied. The OPR ranges from 12 to 36, and the T_{h1} ranges from 1300 to 1600 K. The OPR includes both the fan pressure ratio and the compressor pressure ratio,

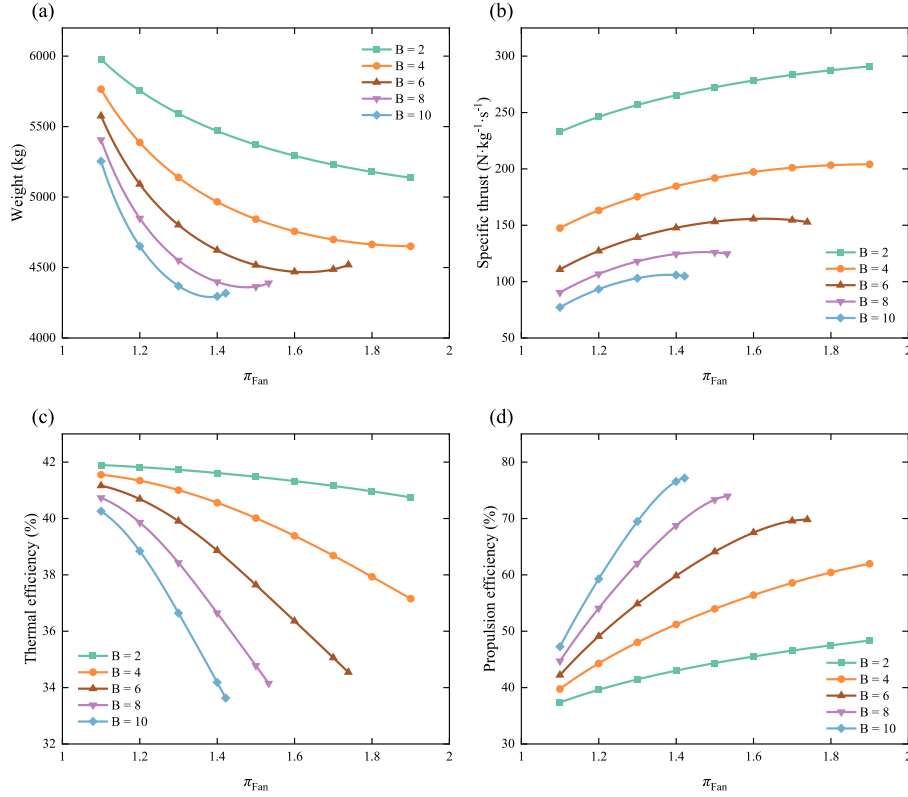


Fig. 11. System performance affected by bypass ratio and fan pressure ratio (a) nuclear power system module weight; (b) specific thrust; (c) thermal efficiency; (d) propulsion efficiency.

which are often studied as a combined parameter during cycle analysis. The impacts of the OPR and the T_{h1} on system performance are illustrated in Fig. 12–Fig. 14. The variations in the weight of the nuclear power module and the thermal power of the reactor are shown in Fig. 12. The thermal power of the reactor and the weight of the nuclear power module increase with the increase of the T_{h1} . Additionally, the weight of the nuclear power module is significantly affected by the temperature difference between the core and the coolant. When the T_{h1} is approximately 1560K, the weight of the nuclear power module reaches the airframe load limitation. Fig. 13 demonstrates the specific thrust, the increase of the T_{h1} has a significant effect on the system

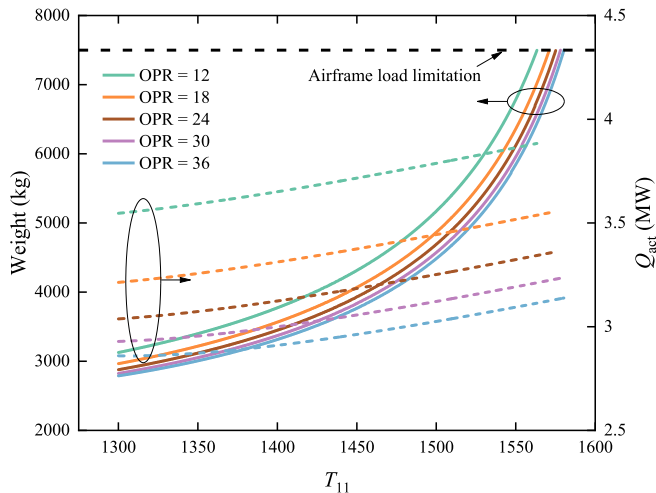


Fig. 12. Nuclear power module weight and thermal power affected by OPR and T_{h1} .

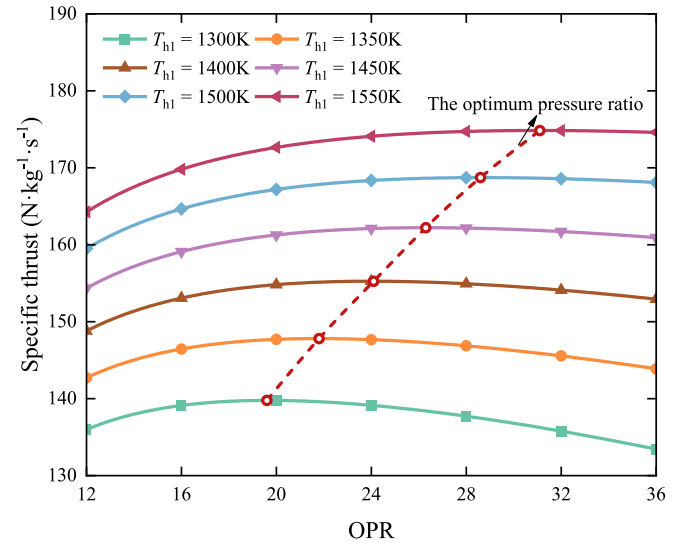


Fig. 13. Specific thrust affected by OPR and T_{h1} .

performance. Furthermore, with the increase of the OPR, there is an optimal value to maximize the specific thrust, which is consistent with the traditional turbine engines. It is worth noting that the optimal OPR increases with the growth of the T_{h1} . However, as seen in Fig. 12, raising the T_{h1} leads to an increase in the weight of the nuclear power module. The thermal efficiency and propulsion efficiency are calculated and given in Fig. 14. The higher OPR and the T_{h1} result in higher thermal efficiency. It is easy to understand that the larger these parameters are, the greater the effective power of the cycle. However, for propulsive efficiency, the influence of the OPR is not as significant as that of the T_{h1} .

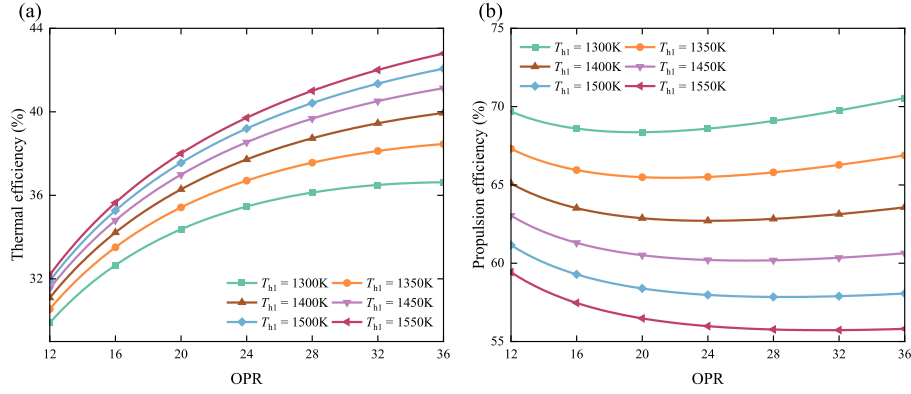


Fig. 14. Thermal efficiency and propulsion efficiency affected by OPR and T_{h1} .

Meanwhile, the lower T_{h1} leads to higher propulsive efficiency.

4.3.4. Influence of OPR and ε_{HE} on system performance

In this section, the effects of overall pressure ratio (OPR) and the effectiveness of heat exchanger (ε_{HE}) on the performance of the hybrid propulsion system are studied, in which the range of OPR is 12–36, the range of ε_{HE} is 0.6–1. Fig. 15 (a) shows that the higher OPR and the lower ε_{HE} , the lower weight of the nuclear power module, albeit at the expense of specific thrust loss. Fig. 15 (b) demonstrates that the higher OPR and the higher ε_{HE} , the higher specific thrust, indicating a significant impact of ε_{HE} on specific thrust. Fig. 15 (c) indicates that the higher OPR and higher ε_{HE} result in higher thermal efficiency. However, for propulsion efficiency, the influence of the OPR is less pronounced compared to that of the ε_{HE} , as shown in Fig. 15 (d).

4.3.5. Multi-objective optimization of system performance

The weight of the nuclear power module and specific impulse are important indicators in the optimization design of hybrid propulsion system. To increase the performance and decrease overall weight, this section employs the Non-Dominated Sorting Genetic Algorithm II (NSGA-II) [1] to optimize the system component parameters.

Considering that long-endurance UAV equipped with hybrid pro-

pulsion system mostly operate during the cruise segment, the optimization plan focuses on the nuclear power stage with the thrust setting of 4.2 kN. Based on the analyses in Sections 4.3.1–4.3.3, the key factors influencing the system performance can be condensed into four parameters: bypass ratio, fan pressure ratio, compressor pressure ratio, and the inlet temperature of the heat exchanger. The compressor and turbine efficiencies are listed in Table 2. The effectiveness of heat exchanger and the pressure recovery coefficient on the air side listed in Table 3. Taking into account the design capability of the turbine components and the power capacity of single-stage turbine, the total temperature ratio between the outlet and inlet of the high-pressure turbine ($T^*_{45}/T^*_{41} \geq 0.73$), and that of the low-pressure turbine ($T^*_{5}/T^*_{45} \geq 0.70$). To maximize the specific impulse and minimize the weight of the nuclear power module, NSGA-II is employed to optimize the above four parameters, which are expressed as:

$$M_{NPM} = \min M_{NPM}(BPR, \pi_{Fan}, \pi_C, T_{h1}) \quad (38)$$

$$F_s = \min F_s(BPR, \pi_{Fan}, \pi_C, T_{h1}) \quad (39)$$

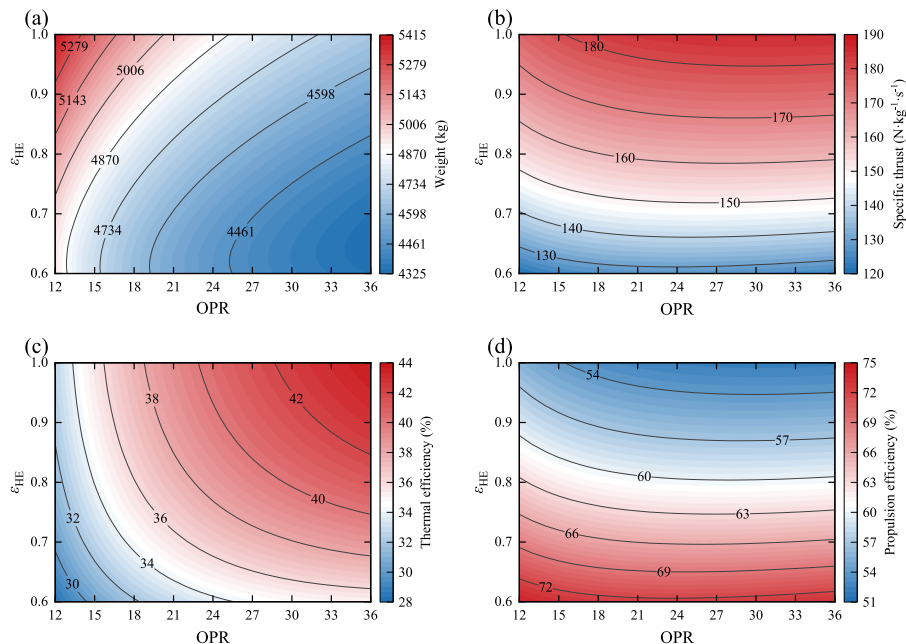


Fig. 15. System performance contour affected by OPR and ε_{HE} .

$$\begin{cases} 4 \leq \text{BPR} \leq 6 \\ 1.2 \leq \pi_{\text{Fan}} \leq 1.8 \\ 12 \leq \pi_c \leq 24 \\ 1450 \leq T_{h1} \leq 1550 \end{cases} \quad (40)$$

The optimal design boundary for the hybrid propulsion system is obtained, as shown in Fig. 16. The optimal range for the weight of the nuclear power module is 3760.1–5788.9 kg, and the optimal range for specific impulse is 150.9–211.7 $\text{N kg}^{-1} \text{s}^{-1}$. The lowest weight of the nuclear power module (3760.1 kg) corresponds to a specific impulse of 150.9 $\text{N kg}^{-1} \text{s}^{-1}$, represented by the leftmost point in Fig. 16. The scenario with the maximum specific impulse (211.7 $\text{N kg}^{-1} \text{s}^{-1}$) corresponds to a weight of the nuclear power module of 5788.9 kg, represented by the rightmost point in Fig. 16. The compromise solution corresponds to the middle position in Fig. 16, with a weight of the nuclear power module of 3941.5 kg and a specific impulse of 195.8 $\text{N kg}^{-1} \text{s}^{-1}$. The detailed optimization results are shown in Table 9.

The scattered distribution plots showing the population dispersion for all the decision variables are illustrated in Fig. 17 (a)–(d). Fig. 17 (a) depicts the scatter plot of the bypass ratio, where the population dispersed around the lower limit in the range of 4–6. While reducing the turbofan engine bypass ratio can enhance the specific impulse while meeting aircraft thrust requirements, it escalates the demand for reactor thermal power, subsequently increasing the weight of the nuclear power module. Fig. 17 (b) displays the scatter plot of the fan pressure ratio, with the population dispersed around the upper limit within the range of 1.2–1.8. Increasing the fan pressure ratio favors the improvement of engine propulsion efficiency, as shown in Fig. 11. Fig. 17 (c) shows the scatter plot of the compressor pressure ratio, with the population dispersed around a pressure ratio of 15. The compressor pressure ratio is restricted by the expansion ratio of the high-pressure turbine. Fig. 17 (d) illustrates the scatter plot of the inlet temperature T_{h1} of the heat exchanger, with the population dispersed around the lower limit within the range of 1450–1550 K. The weight of the nuclear power module is significantly influenced by T_{h1} .

4.4. Flight envelope and weight distribution of hybrid propulsion system

The aircraft flight envelope refers to its capability in terms of Mach number and altitude. When using the nuclear power mode, the turbine inlet temperature is relatively low, which limits the aircraft flight envelope. However, for the nuclear and fossil hybrid propulsion system adopted in this paper, the turbine inlet temperature is higher during burner operation, which can further expand the aircraft flight envelope.

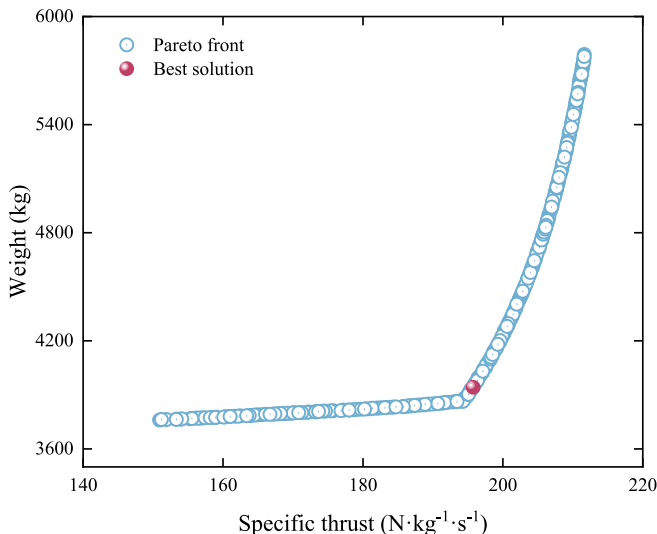


Fig. 16. Two-objective optimization results.

Table 9

Detailed results of two-objective optimization.

Input parameters				Output parameters	
BPR	π_{Fan}	π_c	T_{h1} (K)	Specific thrust ($\text{N} \cdot \text{kg}^{-1} \cdot \text{s}^{-1}$)	M_{NPM} (kg)
4.01	1.79	14.55	1457.7	195.8	3942

After optimization of the hybrid propulsion system, these analyses are detailed in Section 4.4.1. And the analysis of the mass distribution of various components and the fuel consumption for the four flight segments are presented in Section 4.4.2.

4.4.1. Flight envelope

The thrust required to maintain aircraft equilibrium is illustrated in Fig. 6, with a total aircraft weight of 14500 kg. The minimum thrust curve is approximately 4.2 kN. Fig. 18 depicts the flight envelope of the hybrid propulsion system, operating at maximum thrust conditions with a turbine inlet temperature of 1560 K. As flight altitude increases, both the minimum and maximum operating Mach numbers increase. The left boundary of the Mach number is determined by the balance between thrust and drag. When the Mach number is very small, the thrust required to balance the aircraft is large due to the small lift. The hybrid propulsion system cannot be capable of generating such high thrust levels. The right boundary of the Mach number is influenced by the turbine inlet temperature. The peak of the flight envelope is defined by the thrust equilibrium. In regions of high altitude where air density is low, the hybrid propulsion system may not generate sufficient thrust to maintain aircraft equilibrium. The maximum operating altitude is approximately 18.2 km, and the maximum operating Mach number is about 0.71. In the lower right area of the flight envelope, the thrust produced by the hybrid propulsion system surpasses the thrust required for aircraft equilibrium, indicating a significant surplus of thrust.

4.4.2. Weight distribution

The weight distribution of the aircraft is shown in Fig. 19. The maximum take-off weight of the aircraft is 14500 kg, with a fuel capacity of 7500 kg. After implementing the hybrid propulsion system, the carried fuel weight amounts to 2738 kg, occupying 37 % of the available weight. The weights of the turbofan engine, nuclear reactor, ancillary systems, shield, and heat exchanger are 831 kg, 1757 kg, 805 kg, 1264 kg, and 105 kg, representing 11 %, 23 %, 11 %, 17 %, and 1 % of the available weight. Compared to the pre-optimized design, the weight of the nuclear power module in the optimized system is significantly reduced. In the fuel consumption distribution across different segments, 32 % of the fuel is consumed during the climb segment due to the extended climb duration and the need for additional thrust during the initial climb segment. Take-off, Descent, and landing consume a total of 15 % of the fuel. The remaining alternate fuel is 1453 kg, representing 53 % of the carried fuel weight. This alternate fuel can be utilized for maneuvering and acceleration during the mission. For the hybrid propulsion system adopted in this study, the alternate fuel can serve as emergency energy in the event of reactor failure, allowing the UAV to safely return to base.

5. Conclusions

This study demonstrates the integration of nuclear power and fossil fuel in a novel hybrid propulsion system to meet the power requirements of different flight phases and achieve longer endurance. Configuration comparison and performance optimization of the hybrid propulsion system were conducted, and the performance of aircraft equipped with this system was validated. The main conclusions are as follows.

- (1) Due to a higher efficiency, the configuration combining the nuclear power module with a high-BPR turbofan engine becomes the most promising to be equipped on the aircraft, the combined

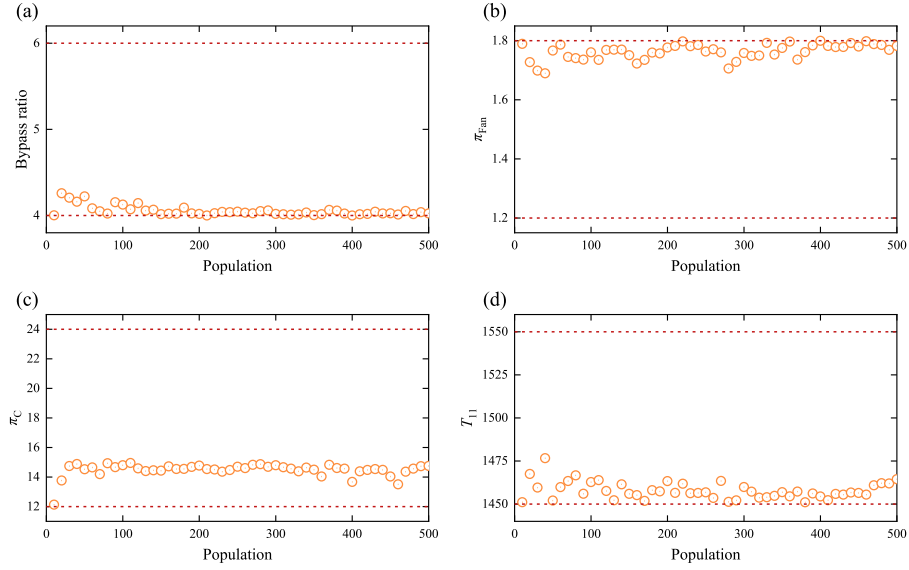


Fig. 17. Optimum range of component parameters.

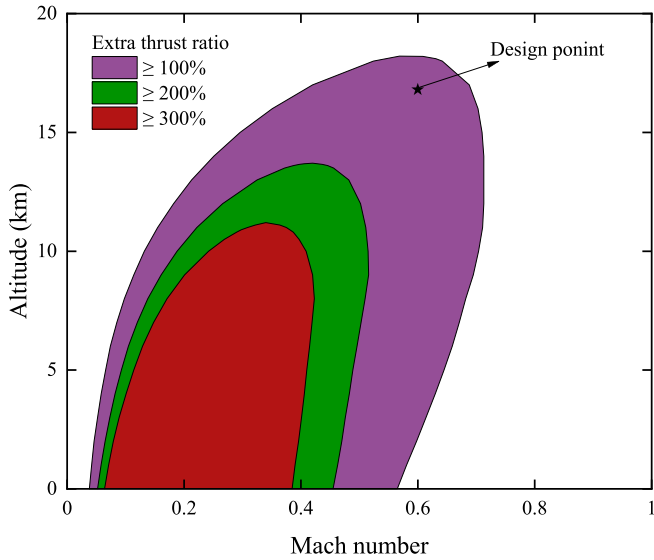


Fig. 18. Flight envelope of the hybrid propulsion system.

component weight and fuel weight of which is 6698 kg, less than the fuel capacity of the original unmodified aircraft;

- (2) In pursuit of the lighter nuclear power module, larger BPR and OPR should be designed, along with the lower T_{h1} ; in pursuit of the higher specific thrust, smaller BPR should be designed, coupled with the higher T_{h1} , where an optimal OPR exists to maximize specific thrust.
- (3) The feature of the Pareto front reveals that the design of system performance is a tradeoff between the nuclear power module weight and the specific thrust: the optimal component parameters can be obtained according to the best solution of the Pareto front; following optimization, the weight of the M_{NPM} is 3942 kg, with the specific thrust of $195.8 \text{ N kg}^{-1} \text{ s}^{-1}$; the corresponding design parameters include the BPR, π_{Fan} , π_C , and T_{h1} , which are 4.01, 1.79, 14.55, and 1457.7 K.
- (4) The aircraft equipped with the hybrid propulsion system operates at the maximum altitude of approximately 18.2 km and reaches a maximum Mach number of 0.71: in the right bottom region, the aircraft exhibits a significant surplus thrust; the components of the hybrid propulsion system occupy part fuel capacity of the original unmodified aircraft, with the turbofan engine, nuclear reactor, ancillary system, shield, and heat exchanger are 11 %, 23 %, 11 %, 17 %, and 1 %.

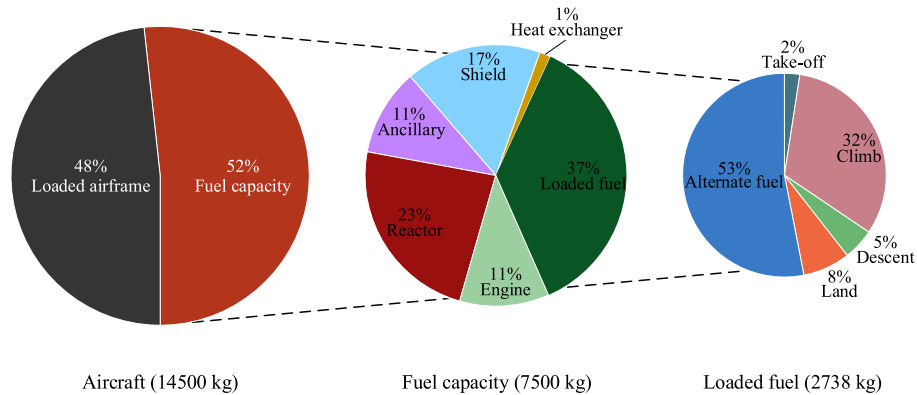


Fig. 19. Weight distribution of the aircraft equipped with the hybrid propulsion system.

The high energy density of the reactor as the aircraft power source, can significantly enhance the endurance of the UAV, but its low power density results in a heavier weight. In our future work, the onboard reactor technology will be one of the key researches to achieve higher power density and safety. Considering the propulsion system composed of many components is structurally complex, the operational reliability will be improved through advanced design methods and testing.

CRedit authorship contribution statement

Kunlin Cheng: Writing – original draft, Investigation, Conceptualization. **Jiahui Li:** Software, Investigation, Data curation. **Fangyan Jiang:** Validation, Software. **Qi Xiao:** Visualization, Software. **Zhenhai Zou:** Visualization, Software. **Jiang Qin:** Writing – review & editing, Conceptualization. **Wuxing Jing:** Writing – review & editing.

Declaration of competing interest

The authors declare there is no conflict of interest regarding the publication of this paper.

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Data availability

The authors do not have permission to share data.

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